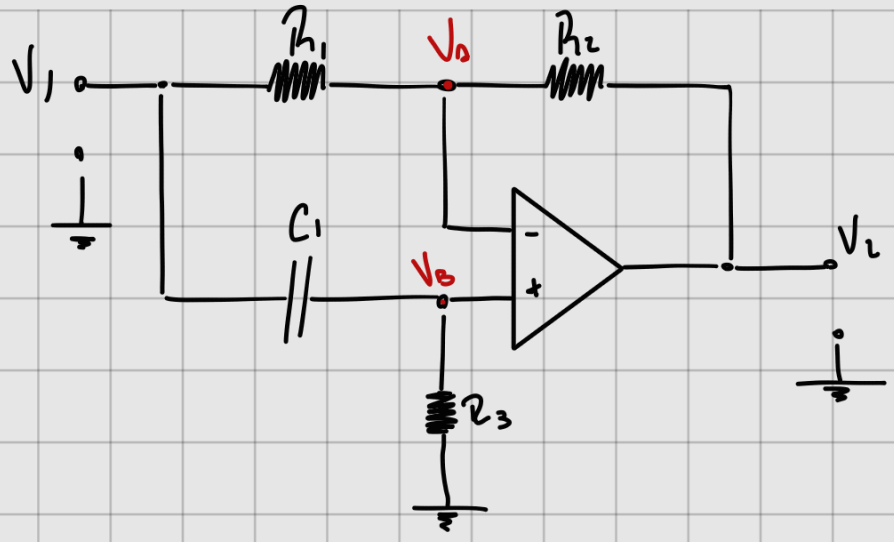


① $V_a(G_1+G_2)-V_2G_2-V_1G_1=\phi$

② $V_a(G_3+sC_1)-V_1(sC_1)=\phi$

② $V_a(G_3+sC_1)=V_1(sC_1)$ $\quad V_a=V_1\left(\dfrac{sC_1}{sC_1+G_3}\right)$

② en ① $V_1\left(\dfrac{sC_1}{sC_1+G_3}\right)(G_1+G_2)-V_2G_2-V_1G_1=\phi \longrightarrow V_1\left(\dfrac{sC_1G_1+sC_1G_2}{sC_1+G_3}\right)-V_2G_2-V_1G_1=0$

$$V_1\left(\frac{sC_1G_1+sC_1G_2}{sC_1+G_3}-G_1\right)=V_2G_2 \longrightarrow V_1\left(\frac{\cancel{sC_1G_1}+sC_1G_2-\cancel{sC_1G_1}-G_1G_3}{sC_1G_2+G_2G_3}\right)=V_2$$

$$V_1\left(\frac{sC_1G_2-G_1G_3}{sC_1G_2+G_2G_3}\right)=V_2 \longrightarrow \frac{V_2}{V_1}=\frac{sC_1G_2-G_1G_3}{sC_1G_2+G_2G_3}$$

$$\frac{V_2}{V_1}=\frac{\cancel{C_1G_2}}{\cancel{C_1G_2}}\cdot\frac{s-G_1G_3/C_1G_2}{s-\frac{\cancel{G_2}G_3}{C_1\cancel{G_2}}} \longrightarrow \boxed{t(s)=\frac{V_2(s)}{V_1(s)}=\frac{s-G_1G_3/C_1G_2}{s-G_3/C_1}}$$

Para el caso en donde el ratio $R_2/R_1=1$

$$\boxed{t(s)=\frac{s-G_3/C_1}{s+G_3/C_1}} \longrightarrow \text{filtro passatodo.}$$

Ceros: G_3/C_1

Polos: $-G_3/C_1$

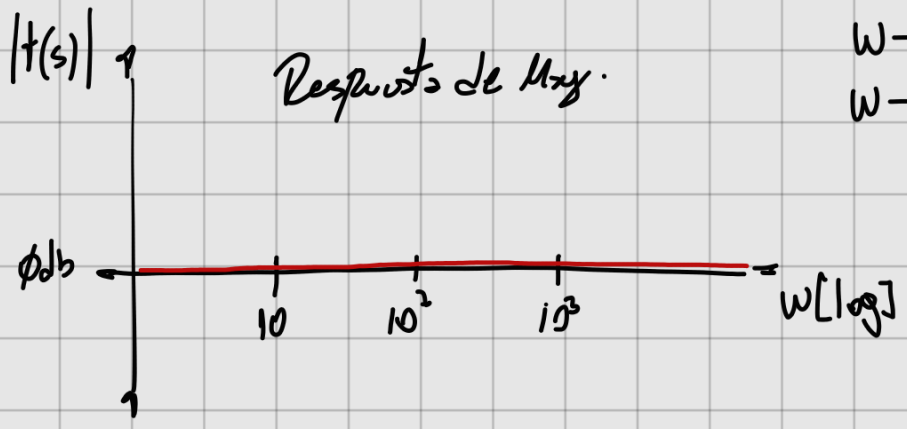
$$|t(j\omega)|=\frac{\sqrt{s^2+(-G_3/C_1)^2}}{\sqrt{s^2+(G_3/C_1)^2}} \longrightarrow \boxed{|t(s)|=1}$$

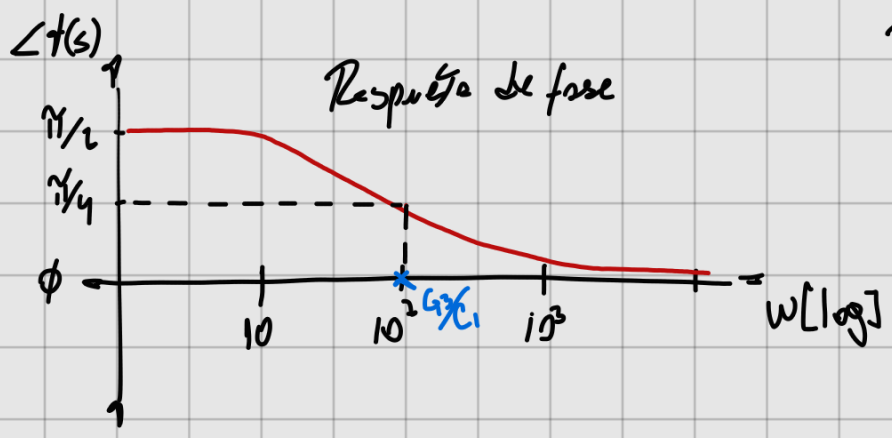
$$\angle t = \underbrace{\angle\left(j\omega - \overbrace{G_3/C_1}^{a}\right)}_{\text{seg cuadrante}} - \angle\left(s^2 + \overbrace{G_3/C_1}^{a}\right)$$

$$\angle t = \pi - \operatorname{arctg}\left(\frac{\omega}{a}\right) - \operatorname{arctg}\left(\frac{\omega}{a}\right) \longrightarrow \angle t = \pi - 2\operatorname{arctg}\left(\frac{\omega C_1}{G_3}\right)$$

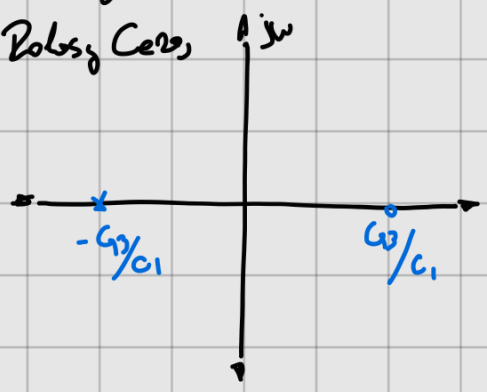
$\omega \to 0: \angle t \to \pi/2$

$\omega \to \infty: \angle t \to 0$





Diagr. Polos y Ceros



Normalización de impedancia.

$$R_3' = \frac{R_3}{\Omega_Z} \quad \text{tomo } \Omega_Z = R_3 \longrightarrow R_3' = 1 \qquad C_1' = C_1 \cdot \Omega_Z.$$

$$t(s) = \frac{s - G_3'/C_1'}{s + G_3'/C_1'} = \frac{s - \frac{G_3 \cdot \cancel{R_3}}{C_1 \cdot \cancel{R_3}}}{s + \frac{G_3 \cancel{R_3}}{C_1 \cancel{R_3}}} = \frac{s - G_3/C_1}{s + G_3/C_1}$$

Normalización de frecuencia

$$\Omega_W = W_0 = \frac{G_3}{C_1} \longrightarrow \$ = \frac{s}{W_0} \qquad t(\$) = \frac{\cancel{G_3/C_1}}{\cancel{G_3/C_1}} \cdot \frac{\$ - 1}{\$ + 1} \qquad t(\$) = \frac{\$ - 1}{\$ + 1}$$

Transferencia Normalizada.